## Location Structure

**building**

| | |
|---|---|
| PK | **buildingID** |
| | buildingName |

**wing**

| | |
|---|---|
| PK | **wingID** |
| FK | *buildingID* |
| | wingName |
| | proximityPool |
| | proximityParking |
| | handicappedAccess |

building has wing

**floor**

| | |
|---|---|
| PK | **foomID** |
| FK | *wingID* |
| | floorNumber |
| | smokingDesignation |

wing has floor

**room**

| | |
|---|---|
| PK | **roomID** |
| FK | *floorID* |
| | roomNumber |
| | baseRate |
| | roomStatus |

floor contains room

## Bed System

**bed_type**

| | |
|---|---|
| PK | **bedTypeId** |
| | size |

bed_type is used in **room_bed**

**room_bed**

| | |
|---|---|
| PK, FK | ***bedTypeId*** |
| PK,FK | ***roomId*** |
| | quantity |

room has room_bed

## Room Adjacency

**room_adjacency**

| | |
|---|---|
| PK, FK | ***roomId1*** |
| PK,FK | ***roomId2*** |
| | hasPrivateDoor |

room_adjacency is adjacent to room

## Charges & Transactions

**service**

| | |
|---|---|
| PK | **serviceId** |
| | serviceType |

service produces **charge**

**charge**

| | |
|---|---|
| PK | **chargeId** |
| FK | *serviceId* |
| FK | *billingPartyId* |
| FK | *roomId* |
| FK | *stayId* |
| FK | *eventId* |
| | amount |
| | chargeDatetime |
| | Description |

notes: roomId is nullable

billing_party pays charge

billing_party incurs charge

room records charge

stay incurs charge

## Billing System

**billing_party**

| | |
|---|---|
| PK | **billingPartyId** |
| FK | *personId* |
| FK | *organizationId* |

**stay**

| | |
|---|---|
| PK | **stayId** |
| FK | *guestId* |
| | checkIn |
| | checkOut |

stay assigned **room_assignment**

**room_assignment**

| | |
|---|---|
| PK, FK | ***stayId*** |
| PK, FK | ***roomId*** |
| PK | **assignedFrom** |
| | assignedTo |

## Reservations & Assignments

**reservation**

| | |
|---|---|
| PK | **reservationId** |
| FK | *billingPartyId* |
| | reservationDate |
| | depositRequired |
| | estimatedGuests |

reservation includes **room_reservation**

**room_reservation**

| | |
|---|---|
| PK, FK | ***reservationId*** |
| PK,FK | ***roomId*** |
| | startDateTime |
| | endDateTime |

room has reservation

reservation makes charge

## Access and Tracking

**access_card**

| | |
|---|---|
| PK | **cardId** |
| FK | *guestId* |
| | pin |

access_card generates **card_access_log**

**card_access_log**

| | |
|---|---|
| PK | **logId** |
| FK | *cardId* |
| FK | *roomId* |
| | accessTime |
| | direction |

stay has card_access_log

## Room Types & Capabilities

**sleeping_room_details**

| | |
|---|---|
| PK, FK | ***roomID*** |
| | *capacity* |
| | smoking |
| | hasToilet |
| | hasTV |
| | hasPhone |

sleeping_room_details contains **suite_details**

**suite_details**

| | |
|---|---|
| PK,FK | **roomID** |
| | sleepingRoomID |
| | meetingRoomID |

**meeting_room_details**

| | |
|---|---|
| PK/FK | **roomID** |
| | seatingCapacity |

meeting_room_details contains suite_details

## People & Roles

**person**

| | |
|---|---|
| PK | **personId** |
| | first_name |
| | last_name |
| | phone |
| | email |

person belongs to **person_organization**

**person_organization**

| | |
|---|---|
| PK,FK | ***personId*** |
| PK,FK | ***organizationId*** |

person_organization associated with **organization**

**organization**

| | |
|---|---|
| PK | **organizationId** |

person is **guest**

**guest**

| | |
|---|---|
| PK,FK | ***guestId*** |

guest has stay

person becomes **host**

**host**

| | |
|---|---|
| PK,FK | ***hostId*** |

host used as person

host organizes **event**

**event**

| | |
|---|---|
| PK | **eventId** |
| FK | *hostId* |
| | startDate |
| | endDate |
| | estimatedAttendance |
| | estimatedGuestRooms |

## Events

event uses **event_room**

**event_room**

| | |
|---|---|
| PK, FK | ***eventId*** |
| PK, FK | ***roomId*** |
| PK | **usageSlot** |